

ons to enhance it to commercial-grade firewalls. A few of the popular IPCop add-ons, and their functions, are listed in Table 1.

Table 1: Add-ons for IPCop v 1.4.21

No	Add-on	Function
1	BlockOut-Traffic (BOT)	Port-by-port blocking of traffic from the internal network to the Internet
2	Zerina	Integrate OpenVPN server functionality in IPCop for remote client connection
3	AdvProxy	To increase functionality of the default proxy available in IPCop
4	URL Filter	Used to block unwanted domains, URLs and files

Installing IPCop with these add-ons converts the basic distribution to a fully functional firewall, which also includes free (and paid) updates for website blacklists. Installing add-ons requires additional configuration work. Administrators used to face various problems while installing, configuring and upgrading the add-ons. In particular, taking back-ups of various add-ons' settings was cumbersome, since IPCop's backup was only for the default settings (without add-ons). Also, new computer hardware, especially most of the network cards, SATA hard disks and flash drives were no longer compatible with IPCop 1.4.21.

IPCop Ver 2.0.x

The latest IPCop release, 2.0.x, addresses these difficulties. It incorporates BOT, Zerina and AdvProxy add-ons. URL Filter is also incorporated, but is expected to be fully functional in release 2.1. The latest release, as of January 1, 2013, is 2.0.6, which also includes drivers for the latest computer hardware and supports installation on flash drives as well.

Installation

Please refer to the June 2012 issue for the installation instructions of IPCop 2.0.4. Start by downloading the ISO image of the latest stable release (2.0.3) from <http://ipcop.org/download.php>. Burn it on a CD. IPCop installation is very simple and straightforward. You require a computer system with a minimum of two Ethernet cards, 512 MB RAM, a hard disk or flash drive, and a CD-ROM drive for installation. Start the installation by booting from the IPCop CD. The first screen greets you with the IPCop mission statement: 'The bad packets stop here'; press *Enter* at the *boot:* prompt. Select the desired language; an information dialogue box tells you that pressing *Cancel* will reboot the system. Continue by clicking *OK*. Select the type of keyboard, the time zone, and enter the correct date and time, if required. Select the hard disk on which to install IPCop. Beware that all data on this disk will be erased. Then click *OK* to continue.

Here, a very interesting screen greets you, irrespective of whether the installation is on the hard disk or on flash. Select

the desired disk type from *Flash/Disk*. The installer will make the required file systems and swap space, ask whether you have an older backup to be restored, and complete the installation.

Reboot the system to continue to configure the box. Enter the host name and domain name, before selecting the Red (Internet) interface type and Ethernet card. Note that here you can configure any Ethernet card to any interface. In the earlier version, the first detected Ethernet was always assigned to Green, and it was tricky to change it later. Also, to identify the card easily, you can set it in 'LED blinking on' mode. Assign Green and Red cards, and enter the Green IP address and subnet mask. Also, select the Red type such as Static, PPPoE, etc. You may enable DHCP if desired, and enter root, admin and backup passwords to complete this part of the configuration.

Box 1: IPCop interface colours

IPCop has given colours to interfaces in an intuitive way. The Green interface represents the trusted internal network, Blue represents wireless or second trusted network, Red represents the Internet which is the most untrusted, and Orange is midway—a DMZ where computer systems that are accessible internally as well as from the Internet are installed.

Configuration

Reboot the system and wait till you see the login: screen (which can be used to log in as the root user). Use a client computer browser to open <https://IPCopIP:8443> and authenticate yourself as the *admin* with the earlier defined password. The first step is to complete the set-up by configuring the Internet. This trial set-up uses PPPoE on the Red interface. Proceed to *Network > Dialup*, and enter the PPPoE username and password. From the *Home* screen (*System > Home*), click the *Connect* button to connect to the Internet. Configure the IPCop Green IP as the default gateway and DNS in the client TCP/IP settings. That is all... you can now start browsing securely.



Figure 1: System Update page